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A potential field based approach to multi robot formation navigation

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Abstract

This paper presents a directed potential field approach for motion co-ordination in formations of multi-robot-systems. First a short introduction to the application of MRS formation navigation is given. The approach is briefly compared to related work. For the sub-task of moving in formation, a solution based on a potential field approach is presented. Different forces belonging to other robots, obstacles and the aspired shape of formation are combined and used to move each robot to its desired position inside the formation. While moving in formation, the group is able to avoid obstacles and approach towards a specified target. First results in simulation and with real robots in the presence of obstacles are presented.

1 Introduction

The idea of multiple robots working together in order to accomplish a task efficiently has become increasingly popular in the last several years. Among these tasks, many situations can be found in which the sub-problem of generating some kind of formation and then maintaining it during a movement must be solved. Possible application areas for multi-robot-systems generating and moving in a formation are

- save and rescue missions,
- sensor and communication networks,
- ordnance disposal, demining and decontamination,
- security applications,
- transport and convoying,
- plant/building management and services,
- agriculture use (e.g. harvesting).

These applications differ in the strictness of the formation constraints. While e.g. a transport mission requires very tight constraints on the robot pattern to share the workload equally among all robots, other applications like re-connaissance and surveillance only need the robots to be spread more or less uniformly over a specified area. Some tasks like convoying or demining build a formation but allow a distortion or partition of the pattern, e.g. while passing obstacles. Under the latter condition four basic subtasks can be identified for a mission that requires moving in a formation:

- assemble at the starting point and/or establish the formation,
- maintain formation while moving,
- avoid obstacles while occasionally splitting/ deforming and then re-establishing the formation,
- change the shape of the formation.

Our main contribution in this paper is the presentation of a directional potential field approach for formation navigation that can cope with those four tasks (section 3) and first results from initial experiments in simulation and with real robots (section 4).

2 Background

Several approaches to the establishing and maintaining subtasks can be found in the literature, which mostly do not address the problem of avoiding obstacles. Examples include papers by Wang [1], who uses neighbour tracking and a mathematical model of the leader-follower co-ordination strategy, and Suzuki et al. [2, 3, 4, 5]. The latter work treat the problem of formation generation in terms of distributed algorithms using idealised robots with perfect sensors and no physical dimension. Chen and Luh [6], like some other authors, improve these algorithms with regard to usefulness for physical robots and propose a method to move a group of mobile robots in strict formation [7]. Yamaguchi and Arai [9] define a potential field spread on the space of relative distances between neighbours, which makes the robots converge into a predefined shape.

Parker [8] discusses the proper balancing between local and global knowledge while moving in a formation. Global information about goals and obstacles should influence the robots local behaviour when avoiding obstacles. Balch and Arkin present a behaviour-based approach to robot formation keeping [10]. Reactive behaviours, implemented as so-called motor schemas,

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correspond to the different influences, which have an effect on the movements of the robots. In [11] this approached is extended by an additional motor schema which is based on a potential field method and which can be used to define uniform and structured "geometric" formations.

Balch's behaviour-based model on the other hand is similar to the constraints our approach is based on. He accepts temporary distortions in a formation in order to avoid obstacles; he also has some kind of relative goal positions for the robots, and makes use of a virtual potential field to move the robots to the desired formation. The fundamental difference is that we assign each robot a special position inside the formation whereas Balch wants the robots to be randomly distributed over the possible formation positions. Additionally he can only design special uniformly structured formations because the possible relative positions (which he calls "attachment sites") are the same for each robot. In our approach, it is simple to establish any desired shape of formation.

Song and Kumar [17] present a potential field approach where the robot group has to get into a certain formation at a specified goal point. It is not clear if the formation can move as a whole. Only static obstacles are assumed. Contrary to our approach a single robot cannot have a specified location in the formation.

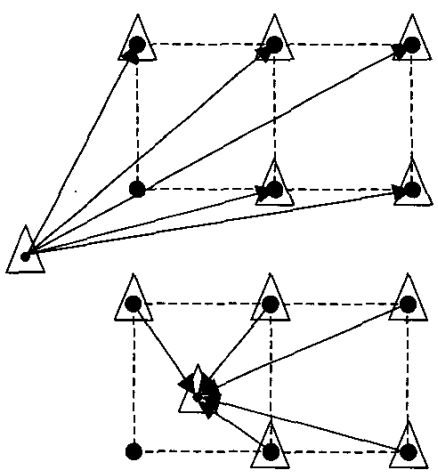

Figure 1. Robot outside (top) and inside (bottom) the formation.

3 Potential field

If strict formation constraints are not required and deforming of the formation in the presence of obstacles is allowed, two opposite influences on the resulting paths for each robot can be identified: reaching a goal point on the shortest possible path and maintaining the formation.

Obviously, these two demands contradict in many environments, e.g. at doorways or in narrow corridors. Without distortion of the formation a detour might be necessary. In order to model this situation we use a potential field approach as described in [13] and therefore assign each robot a specific position inside the formation. As an extension to the similar method presented in [9] not only a shape-forming force is used to generate the formation, but also additional types of forces are modelled and combined to achieve more functionality and flexibility.

The concept of using a potential field method for controlling a formation is based on the view that not only obstacles and goals can exert forces onto a robot [14, 15]. The basic idea is that each robot in a formation can exert repulsive and attractive forces to other robots. These virtual forces can be used to keep the robots in the desired formation.

The example in Figure 1 shows a robot that is outside the formation. The attractive forces of the other robots will "pull" it back into the formation. On the other hand, the attractive force of the robot itself will cause the formation to move towards its position. If a robot has moved too close to the formation, the repulsive forces of the other robots will "push" it away. In both cases, the forces will move the robot and the formation into a constellation where the desired formation is re-established.

Two forces have to be considered when the formation is forming up or is in motion: the repulsive obstacle forces and the forces that establish and maintain the formation. The following two sections will discuss these different forces more in detail.

3.1 Formation forces

Let $p_i \in \mathbb{R}^2$, $i = 1,\ldots,n$ be the positions of n robots in the formation and $o_i \in [0;2\pi)$, $i = 1,\ldots,n$ the orientation in the target formation, both in an arbitrary Cartesian coordinate system T_F. Independent of the configuration of T_F in a world coordinate system T_W, each robot k expects the other robot at a certain position seen from his orientation. Under the assumption that the desired goal position and orientation has been reached, the expected goal position $\hat{p}_{k,i}$ of robot i seen from robot k is:

$$\hat{p}_{k,i} = p_k + \begin{pmatrix} \cos(o_{ist} - o_i) & -\sin(o_{ist} - o_i) \\ \sin(o_{ist} - o_i) & \cos(o_{ist} - o_i) \end{pmatrix} (p_i - p_k) \quad (1)$$

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where o_{ist} is the current orientation of robot k.

To determine the potential field that arranges the robot group in the desired formation, for each robot k the positions where the other $n-1$ robots expect robot k are defined by $U_{form,j}:\mathbb{R}^2 \to \mathbb{R}$ with

$$U_{form,j}(x)=\frac{1}{2}\xi\left\|x-\hat{p}_{j,k}\right\|^2,$$

$j=1,\ldots,k-1,k+1,\ldots,n$, and ξ a constant scaling factor[1].

This results for robot k in the formation potential $U_{form}:\mathbb{R}^2 \to \mathbb{R}$,

$$U_{form}(x)=\sum_{\substack{j=1\\ j\neq k}}^{n} U_{form,j}(x).$$

Accordingly the force $F_{form}:\mathbb{R}^2 \to \mathbb{R}^2$, that applies to another robot k at the position x *is*

$$F_{form}(x)=-\nabla U_{form}(x),$$

which results in

$$F_{form}(x,y)=\begin{pmatrix} -\frac{\partial}{\partial x}\sum_{\substack{j=1\\ j\neq k}}^{n} U_{form,j}(x,y) \\ -\frac{\partial}{\partial y}\sum_{\substack{j=1\\ j\neq k}}^{n} U_{form,j}(x,y) \end{pmatrix}^T$$

$$=\begin{pmatrix} -\sum_{\substack{j=1\\ j\neq k}}^{n} \frac{\partial}{\partial x}\frac{1}{2}\xi\left\|(x,y)-\hat{p}_{j,k}\right\|^2 \\ \sum_{\substack{j=1\\ j\neq k}}^{n} \frac{\partial}{\partial y}\frac{1}{2}\xi\left\|(x,y)-\hat{p}_{j,k}\right\|^2 \end{pmatrix}^T$$

$$=\begin{pmatrix} -\sum_{\substack{j=1\\ j\neq k}}^{n} \frac{\partial}{\partial x}\frac{1}{2}\xi\left[\left(x-\hat{x}_{j,k}\right)^2+\left(y-\hat{y}_{j,k}\right)^2\right] \\ \sum_{\substack{j=1\\ j\neq k}}^{n} \frac{\partial}{\partial y}\frac{1}{2}\xi\left[\left(x-\hat{x}_{j,k}\right)^2+\left(y-\hat{y}_{j,k}\right)^2\right] \end{pmatrix}^T$$

$$=\xi\left(\sum_{\substack{j=1\\ j\neq k}}^{n}\left(\hat{x}_{j,k}-x\right),\sum_{\substack{j=1\\ j\neq k}}^{n}\left(\hat{y}_{j,k}-y\right)\right)$$

[1] Notice. The indices j and k are swapped compared to the definition of $\hat{p}_{k,i}$ in (1).

where $\left(\hat{x}_{j,k},\hat{y}_{j,k}\right):=\hat{p}_{j,k}\in\mathbb{R}^2$ and $\hat{p}_{k,i}$ are used as in (1). Note that the transposition in the intermediate steps is used only to stick to the one column size.

Simply worded, for every robot the goal position to (re-) establish the desired formation with respect to the positions of the other robots is calculated. The vector between the current position and the formation position is transformed into an attractive or a repulsive force.

3.2 Obstacle and goal forces

In addition to the formation forces, a repulsive force caused by obstacles (or maybe by other robots) and, in this version of the implementation, a goal force that attracts the formation to a certain point must also be taken into account.

Let $H\subset\mathbb{R}^2$ be the set of all obstacles, so for all points $x\in\mathbb{R}^2-H$ outside of obstacles the repulsive potential is $U_{rep}:\mathbb{R}^2\to\mathbb{R}$

$$U_{rep}(x)=\begin{cases}\frac{1}{2}\eta\left(\frac{1}{\delta(x)}-\frac{1}{\delta_0}\right)^2 & \text{, if } \delta(x)\le\delta_0\\ 0 & \text{, if } \delta(x)>\delta_0.\end{cases}$$

where

$$\delta(x):=\min_{x'\in H}\left\|x-x'\right\|$$

is the minimal distance to all obstacles and δ_0 the maximum influence of the obstacles. For the repulsive force $F_{rep}:\mathbb{R}^2\to\mathbb{R}^2$, that affects robot k of a group on position x, correspondingly applies $F_{rep}(x)=-\nabla U_{rep}(x)$, which results in

$$F_{rep}(x)=\begin{cases}\left(-\frac{\partial}{\partial x}U_{rep}(x),-\frac{\partial}{\partial y}U_{rep}(x)\right) & \text{, if } \delta(x)\le\delta_0\\ 0 & \text{, if } \delta(x)>\delta_0\end{cases}$$

$$=\begin{cases}\eta\left(\frac{1}{\delta(x)}-\frac{1}{\delta_0}\right)\frac{1}{\delta^2(x)}\nabla\delta(x) & \text{, if } \delta(x)\le\delta_0\\ 0 & \text{, if } \delta(x)>\delta_0\end{cases}$$

In the current implementation, we use a simplified approach for the repulsive forces. The method for generating the repulsive forces is a modified approach similar to the one proposed by Borenstein and Koren [14]. One of the major variations is that no certainty grid is

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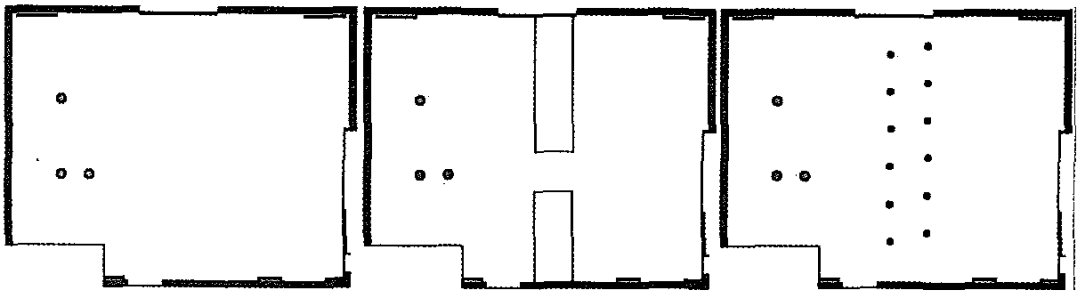

Figure 2. Experimental settings: Empty hall (f), doorway (d) and gallery (g). The dots on the left are 3 robots.

used. Because only local obstacle avoidance[2] is carried out a map is not necessary. The sensor readings are basically transformed into the corresponding repulsive forces and add up. This is of course not mathematically correct, but gives a useable approximation.

The approach of calculating the goal force is similar to the one for the formation force. Let $z := (x_z, y_z) \in \mathbb{R}^2$ be the desired goal position. Then the corresponding potential field is defined by $U_{att} : \mathbb{R}^2 \to \mathbb{R}$,

$$U_{att}(x) = \frac{1}{2}\zeta \|x - z\|^2,$$

with a positive scaling constant $\zeta \in \mathbb{R}$. For the resultant attracting force $F_{att} : \mathbb{R}^2 \to \mathbb{R}^2$, which affects a robot in position x, holds $F_{att}(x) = -\nabla U_{att}(x)$, that is

$$\begin{aligned} F_{att}(x,y) &= \left(-\frac{\partial}{\partial x} U_{att}(x,y), -\frac{\partial}{\partial y} U_{att}(x,y) \right) \\ &= \zeta (x_z - x, y_z - y) \end{aligned}$$

The overall resulting force $F(x) : \mathbb{R}^2 \to \mathbb{R}^2$, which affects a robot at position x, therefore is

$$F(x) = F_{form}(x) + F_{rep}(x) + F_{att}(x).$$

4 Initial Experiments

In order to validate that the presented method of a directed potential field can be transferred into a real world environment, first experiments with simulated and real robots were performed. Thus, the main objective was to compare the driven paths of the mathematical model that uses free-floating objects as robots with a more realistic robot simulator [16] and finally with our real robots. Thereby it is assumed that the workspace of the robot group consists of both moving and stationary obstacles with arbitrary shapes. The group of robots has an initial formation and position. A different goal formation for the robots is specified. The goal of the experiments was to let the robots move towards some target location and position in their goal formation simultaneously, while at the same time avoiding obstacles and collisions with other robots.

Three different experimental sets have been used, first a simulation based on the pure mathematical model, second a multi robot simulation, and third experiments with a real B21 and several Pioneer I robots. There were also three different environments, first an empty space (f), second a doorway (d), and third a kind of gallery (g) that had to be passed (see fig. 2). In all three cases the starting positions were in the left part and the desired target location for the formation was the lower right corner of the hall. We conducted experiments with two robots and with three robots, for the two robots the goal formation was to move side by side, for three robots it was a triangle formation. For each combination of these settings three different runs were performed and the resulting paths were recorded.

The upper chart in figure 3 shows the deviation between the mathematical model and the paths generated by the multi robot simulation. For each setting the mean deviance for all robots from the optimal formation position as calculated in the mathematical model is shown. In spite of the small number of only three runs the standard deviation was added for completeness. As one can see, the simulation runs are quite accurate, even for the three robot example. The only mentionable discrepancy is the gallery setting with two robots (g2) which is even worse than the three robot version. We are not yet quite sure about the reasons, but the fact is subject to further investigations.

[2] Since this paper is focused on the formation problem under consideration of local obstacles higher path planning problems are beyond the scope of this paper. We will not use a potential field approach for global path planning, just for the experiments!

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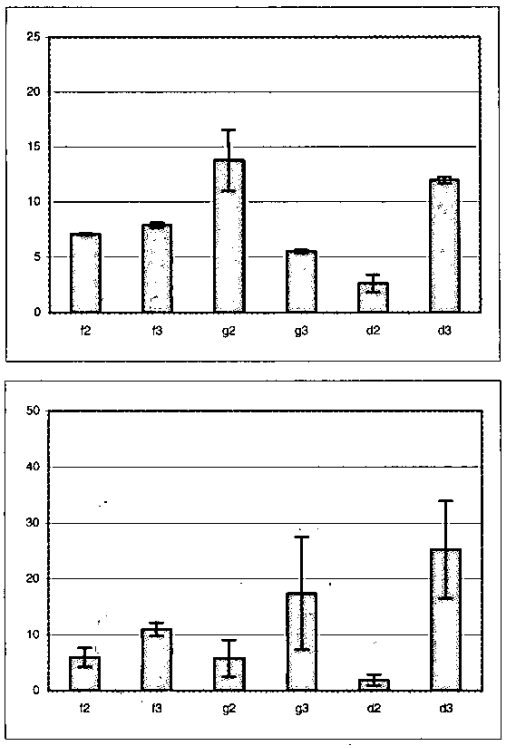

Figure 3. Track deviation for simulator (top) and for real robots (bottom).

The other chart presents the deviation of the real robots compared to the simulated ones. Obviously, the differences in the two robot settings are also rather small. With the use of three robots, results vary strongly which can bee seen from the large standard deviation. Apart from the fact that in our simulation robots simply move in a more fluent and precise manner than their real counterparts, there might be another important reason. We used a B21 and two Pioneer I for this experiment. It turned out that the B21 accelerates, turns and moves much faster than the Pioneer I. One way to cope with this might be to restrict the acceleration and velocity of all robots to the same limit. The more elegant way, of course, would be to extend the 3 dimensional planning algorithm (x, y, θ) with an acceleration or velocity dimension. This could be done by using the magnitude of the formation force.

Overall, these initial experiments showed that the developed navigation method for formations of multi-robot-systems works in an simplified simulation environment as well as with real robots. As first results can be mentioned that

- in real environments the robots nearly behave like in the simulation,
- because of its simplicity the method is fast enough to work in "real time", and
- the approach is flexible concerning the shape of the formation and the number of robots.

5 Conclusion and Future Work

In this paper, a method for navigating formations of multiple mobile robots was presented. Because in our context a formation is not regarded as strict but as capable of deforming and splitting in the presence of obstacles, we make use of a virtual force field in order to establish and maintain a particular formation. In our view, obstacles and goal points as well as other robots belonging to the same formation exert virtual forces. These different types of forces, which influence the resulting paths for each robot, are modelled and combined, including attractive and repulsive shape forces and obstacle forces, in order to generate steering commands to move the robot.

The first experiments with simulation and real robots where quite promising. The forming of formations with and without the presence of obstacles produced stable results for various shapes of formation and different numbers of robots. The initial experiments proofed the results of the mathematical model. The method is flexible and stable enough to cope with the demands of "real robots".

When the robots and other moving obstacles change position, the potential fields are updated accordingly. Regardless of the number of robots in the group and the sizes of the obstacles, the robots avoid colliding with each other or with obstacles, and they reach their goal point successfully. The validity of the method has been verified in simulation and experimentally with real robots.

Of course, already these preliminary experiments revealed some problems and several word to do. Our major points of interest for future experiments are:

- the use of the magnitude of the formation forces to adjust acceleration and velocity,
- tests with higher numbers of robots and a wider variety of shapes,
- a comparison of our directed approach with other methods like Balch's [10, 11] or Song's [17],
- the introduction of tracking in order to improve planning and parameter adjustment.

Another topic which has not been addressed so far is the fact that no global path planning has been considered. The mentioned goal force was used for experimental purposes only. In order to navigate through more complicated environments an additional global path planning for the

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whole formation of robots is necessary. This is a complex task, at least if it is not restricted to the piano moving problem in which formation are not allowed to change shape or to deform. We are currently working on possible combinations of the presented potential field approach with other path planning techniques.

6 References

[1] P. K. C. Wang, *Navigation Strategies for Multiple Autonomous Mobile Robots Moving in Formation*, Journal of Robotic Systems 8(2), 1991, pp. 177-195.

[2] K. Sugihara and I. Suzuki, *Distributed Motion Coordination of Multiple Mobile Robots*, Proc. of 5th IEEE International Symposium on Intelligent Control, Philadelphia, September 1990, pp. 138-143.

[3] I. Suzuki and M. Yamashita, *Formation and Agreement Problems for Anonymous Mobile Robots*, Proc. of 31th Annual Allerton Conference on Communication, Control and Computing, University of Illinois, Urbana, September 1993, pp. 93-102.

[4] I. Suzuki and M. Yamashita, *A Theory of Distributed Anonymous Mobile Robots – Formation and Agreement Problems*, Technical Report TR-94-07-01, Department of Electrical Engineering and Computer Science, University of Wisconsin, Milwaukee, Juli 1994.

[5] H. Ando, I. Suzuki, M. Yamashita, *Formation and Agreement Problems for Synchronous Mobile Robots with Limited Visibility*, Proc. of International Symposium on Intelligent Control, Monterey, August 1995, pp. 453-460.

[6] Q. Chen and J. Y. S. Luh, *Distributed Motion Coordination of Multiple Robots*, Proc. of International Conference on Intelligent Robots and Systems, Federal Armed Forces University Munich, September 1994, pp. 1493-1500.

[7] Q. Chen and J. Y. S. Luh, *Coordination and Control of Small Mobile Robots*, Proc. of 11th IEEE International Conference on Robotics and Automation, San Diego, May 1994, pp. 2315-2320.

[8] L. E. Parker, *Designing Control Laws for Cooperative Agent Teams*, Proc. of 10th IEEE International Conference on Robotics and Automation, Atlanta, May 1993, pp. 582-587.

[9] H. Yamaguchi and T. Arai, *Distributed and Autonomous Control Method for Generating Shape of Multiple Mobile Robot Group*, Proc. of International Conference on Intelligent Robots and Systems, Federal Armed Forces University Munich, September 1994, pp. 800-807.

[10] T. Balch and R. C. Arkin, *Behavior-based Formation Control for multiagent robot Teams*, IEEE Transactions on Robotics and Automation, 1999.

[11] T. Balch and M. Hybinette, *Social Potentials for Scalable Multi-Robot Formations*, Proc. of the 2000 IEEE Int. Conference on Robotics and Automation, San Francisco, April 2000, pp. 73-80.

[12] G. Beni, *The Concept of Cellular Robotic Systems*, Proc. of 3rd IEEE International Symposium on Intelligent Control, Arlington, August 1988, pp. 57-62.

[13] Y. Koren and J. Borenstein, *Potential Field Method and Their Inherent Limitations for Mobile Robot Navigation*, Proc. of 8th IEEE International Conference on Robotics and Automation, Sacramento, April 1991, pp. 1398-1404.

[14] J. Borenstein and Y. Koren, *Real-time Obstacle Avoidance for Fast Mobile Robots*, IEEE Transactions System, Man and Cybernetics, Vol. 19, No. 5, Sept./Oct. 1989, pp. 1179-1187.

[15] J. Borenstein and Y. Koren, *Real-time Obstacle Avoidance for Fast Mobile Robots in Cluttered Environments*, Proc. of 7th IEEE International Conference on Robotics and Automation, Cincinnati, Ohio, May 1990, pp. 572-577.

[16] F. E. Schneider and D. Wildermuth, *Simulation von Mehrrobotersystemen*, 13. Fachgespräch Autonome Mobile Systeme (AMS), Stuttgart, Oktober 1997.

[17] P. Song and V. Kumar, *A Potential Field Based Approach to Multi-Robot Manipulation*, Proc. of the 2002 IEEE Int. Conference on Robotics and Automation, Washington, May 2002, pp. 870-876.

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